Executive Report: Bayesian Network Modelling and Probabilistic Inference for Heart Disease Prediction

EDAB 6808 Assignment 3

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1 Executive Summary

The purpose of this assignment is to build, compare, and evaluate two Bayesian networks for heart-disease prediction: one network specified using domain-informed reasoning and one network learned from data using `bnlearn`. The analysis follows the complete Bayesian-network workflow required by the assignment rubric: data preparation, structure learning, parameter learning, probabilistic inference, and held-out model evaluation.

The supplied Cleveland dataset contains 303 observations and 13 predictive variables, with the original diagnostic target ranging from 0 (no heart disease) to 4 (presence of disease at varying severity). Consistent with the standard binary use of the Cleveland dataset, the target was converted to 0 for no disease and 1 for any original value from 1 to 4 (UCI Machine Learning Repository 1988; Detrano *et al.* 1989). The final binary target was reasonably balanced: 164 observations (54.1%) represented no disease and 139 (45.9%) represented disease. Four encoded missing values in `ca` and two in `thal` were identified and handled using training-derived mode imputation.

The expert DAG contained 16 directed edges and four direct parents of the target, while the data-driven DAG contained 11 directed edges and one direct parent of the target. The networks shared three directed edges and five adjacencies when direction was ignored, but two relationships were reversed. Importantly, the learned structure was interpreted as a probabilistic dependency representation rather than a causal medical model.

The inference stage evaluated 5,184 combinations of clinically relevant variables for each network. The highest posterior probability of heart disease was approximately 0.793 in the expert DAG and 0.670 in the learned DAG. Across all evidence profiles, the mean absolute difference between the models' posterior probabilities was 0.056, with a Spearman rank correlation of 0.766. This indicates substantial, although incomplete, agreement in how the two networks rank risk profiles.

On the held-out test set, the expert DAG outperformed both the data-driven DAG and the majority-class baseline. The expert model achieved 90.2% accuracy, an F1-score of 0.893, ROC-AUC of 0.961, and PR-AUC of 0.953. The data-driven DAG achieved 85.3% accuracy, an F1-score of 0.836, ROC-AUC of 0.915, and PR-AUC of 0.909. The majority baseline achieved only 54.1% accuracy and 0.500 balanced accuracy.

The principal recommendation is therefore to retain the expert DAG as the preferred model for this assignment because it combines stronger held-out performance with explicitly justified pathways. This recommendation is limited to the current dataset and split; the network should not be interpreted as a clinically validated diagnostic system.

2 Introduction and Research Objective

Bayesian networks are probabilistic graphical models that represent a joint probability distribution through a directed acyclic graph and a set of local conditional probability distributions (Heckerman 1995; Koller and Friedman 2009). Their value in analytical decision-making is that they provide both a structural representation of dependencies and a formal mechanism for updating probabilities when new evidence is observed.

This assignment applies Bayesian networks to the Cleveland heart-disease dataset, which contains

303 observations and a widely used subset of clinical and diagnostic attributes (UCI Machine Learning Repository 1988; Detrano *et al.* 1989). The original target is integer-valued from 0 to 4. The modelling task therefore focuses on the binary question of whether heart disease is absent or present.

The central research question is:

> *How do an expert-designed Bayesian network and a data-driven Bayesian network differ in structure, probabilistic inference, predictive performance, and interpretability when applied to binary heart-disease prediction?*

3 Data Collection and Pre-Processing

3.1 Dataset and target definition

The dataset contains 303 observations, 13 predictive features, and one heart-disease outcome. The supplied CSV labels the binary sex variable as `gender`; this report retains that source-column name while recognising that it corresponds to the `sex` variable described in the assignment and UCI documentation.

The original outcome was converted to a binary target:

$$\texttt{heartdisease_bin} = \begin{cases} 0, & \text{if original heartdisease} = 0, \\ 1, & \text{if original heartdisease} \in \{1, 2, 3, 4\}. \end{cases}$$

Table 1: Data-preparation summary

Item	Final analytical treatment
Raw observations	303 patients
Predictive variables	13 clinical/demographic variables
Binary target	0 = no disease; 1 = disease present
Target balance	164 no disease (54.1%); 139 disease (45.9%)
Encoded missing values	4 in `ca`; 2 in `thal`
Missing-value treatment	Training-derived mode imputation: `ca`=0, `thal`=3
Train/test split	242 training observations; 61 held-out test observations
Train/test disease prevalence	Approximately 45.9% in both sets due to stratification
Continuous variables discretised	age, trestbps, chol, thalach, oldpeak
Discretisation strategy	Three quantile-based states learned on training data only

The binary target is not severely imbalanced, which reduces the risk that accuracy alone will be dominated by the majority class. Nevertheless, balanced accuracy, recall, specificity, ROC-AUC, and PR-AUC were retained for a more complete evaluation.

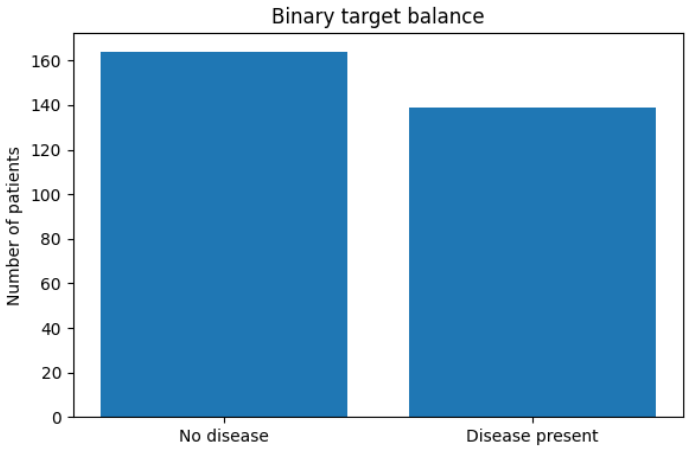


Figure 1: Binary target balance

*T*he original diagnostic values 1-4 were grouped into the "disease present" class, producing 164 no-disease observations and 139 disease observations.

3.2 Exploratory analysis and data quality

The initial schema validation confirmed 303 rows, no duplicates, and the expected coded states for the categorical variables after the encoded missing values were converted to true missing values.The affected variables , ca with 4 missing values and thal with 2 missing values, were then imputed using the mode computed on the training set only, to prevent information from the test set leaking into preprocessing . The continuous/ordered measurements showed the strongest pairwise descriptive relationships between the target and `oldpeak` ($r \approx 0.42$), and between the target and `thalach` ($r \approx -0.42$). These correlations were treated as descriptive rather than causal evidence.

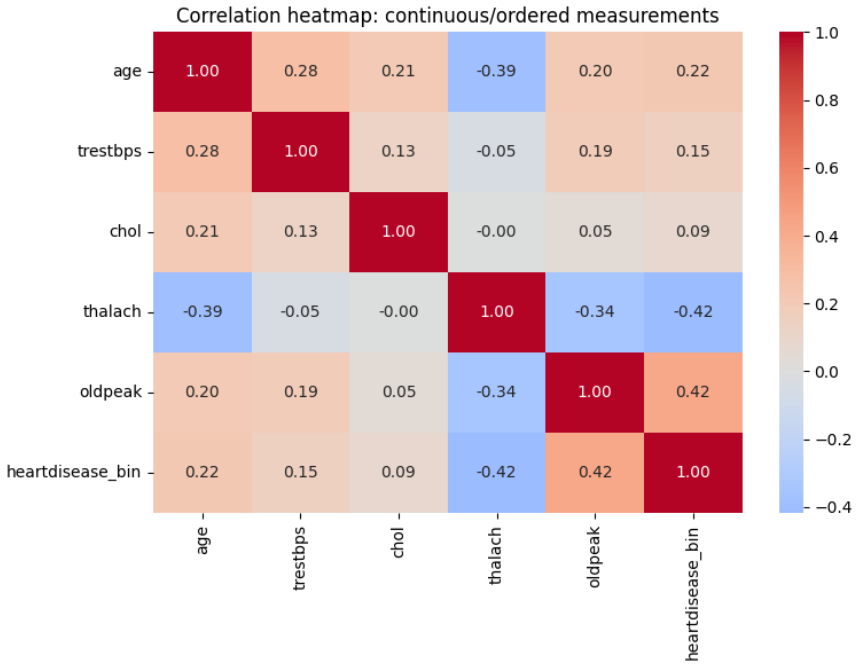


Figure 2: Correlation heatmap for continuous and ordered measurements

Note. Pearson correlations are shown only for the numeric/ordered measurements used in the exploratory stage. The heatmap is descriptive and is not used as evidence of causal direction in either Bayesian network.

3.3 Train/test separation and discretisation

The data were split before imputation and discretisation. This ordering is important because it prevents information from the held-out test set from influencing preprocessing decisions. Missing values were imputed using modes computed from the training set only, after which the same values were applied to the test set.

Five continuous variables were converted to Low, Medium, and High states using training-derived quantile boundaries. The same cut-points were then applied unchanged to the held-out test set.

Table 2: Training-derived discretisation intervals

Raw variable	BN variable	State	Interval
age	age_cat	Low	$(-\infty, 51]$
		Medium	$(51, 59]$
		High	$(59, \infty)$
trestbps	trestbp_cat	Low	$(-\infty, 120]$
		Medium	$(120, 139.333]$
		High	$(139.333, \infty)$
chol	chol_cat	Low	$(-\infty, 226]$
		Medium	$(226, 266.667]$
		High	$(266.667, \infty)$
thalach	thalach_cat	Low	$(-\infty, 143]$
		Medium	$(143, 162]$
		High	$(162, \infty)$
oldpeak	oldpeak_cat	Low	$(-\infty, 0.1]$
		Medium	$(0.1, 1.2]$
		High	$(1.2, \infty)$

The use of discrete states is consistent with the discrete structure- and parameter-learning workflow implemented by the Python `bnlearn` package (Taskesen 2024).

4 Expert DAG and Data-Driven Structure Learning

4.1 Expert-designed DAG

The expert network was constructed as a DAG with 16 directed edges. Rather than connecting nearly every variable directly to the disease target, the final network uses a more compact structure with four direct target parents: `age_cat`, `gender`, `trestbp_cat`, and `chol_cat`. This results in only 54 possible direct-parent configurations for the target, making the target CPT substantially more estimable from 242 training observations.

The expert graph was organised into three reasoning pathways:

1. **Risk-profile pathway:** age, gender, resting blood pressure, and cholesterol feed directly into the target.
2. **Diagnostic-evidence pathway:** chest pain, resting ECG, exercise-induced angina, `ca`, `thal`, slope, and maximum heart rate are represented as downstream findings whose observed states can still update the target posterior through Bayesian inference.
3. **Exercise-response pathway:** age, exercise-induced angina, slope, maximum heart rate, and oldpeak are connected through additional conditional relationships.

The arrows define the model factorisation and are not asserted to be verified physiological causal effects.

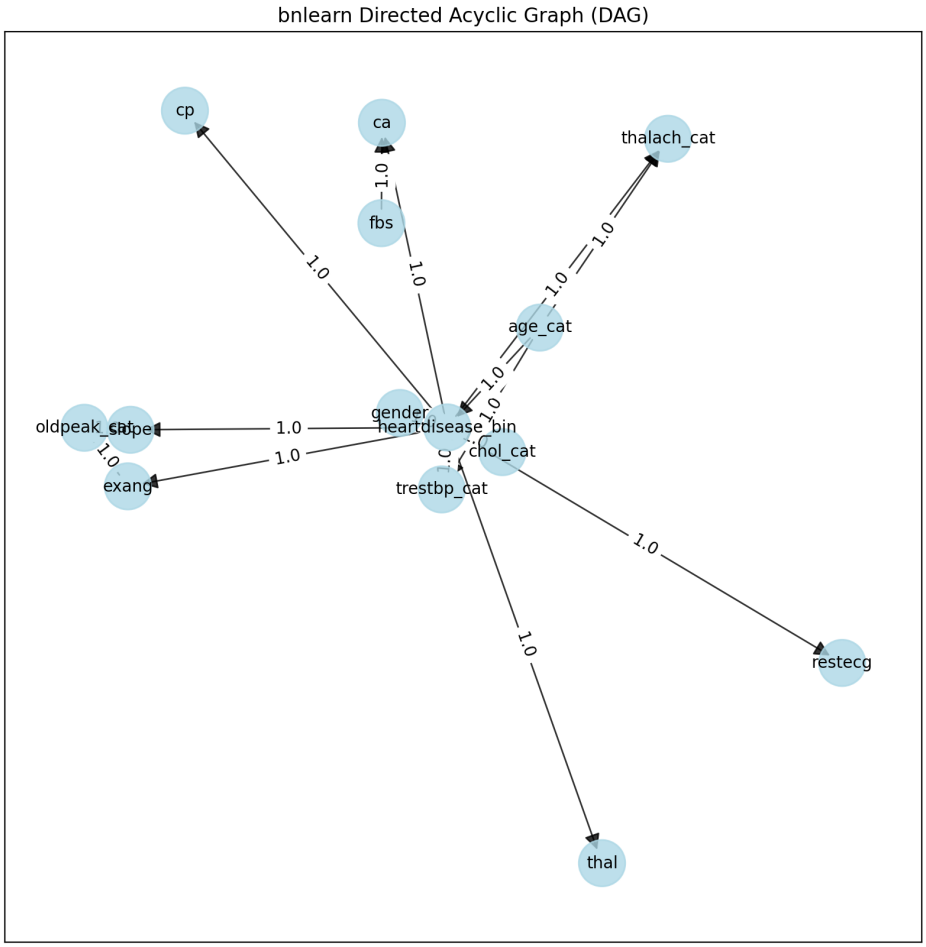


Figure 3: Expert-designed Bayesian network

Note. The image is the `bnlearn` rendering produced by the notebook. Although the plotting function uses the generic title "bnlearn Directed Acyclic Graph (DAG)", this figure represents the manually specified expert DAG. Arrows encode the chosen probabilistic factorisation, not established medical causality.

4.2 Data-driven DAG

The second structure was learned from the training data using Hill Climbing with the Bayesian Information Criterion (BIC). Hill Climbing performs a local search over possible edge additions, deletions, and reversals, while BIC rewards model fit and penalises unnecessary structural complexity (Schwarz 1978). The final learned DAG contained 11 directed edges.

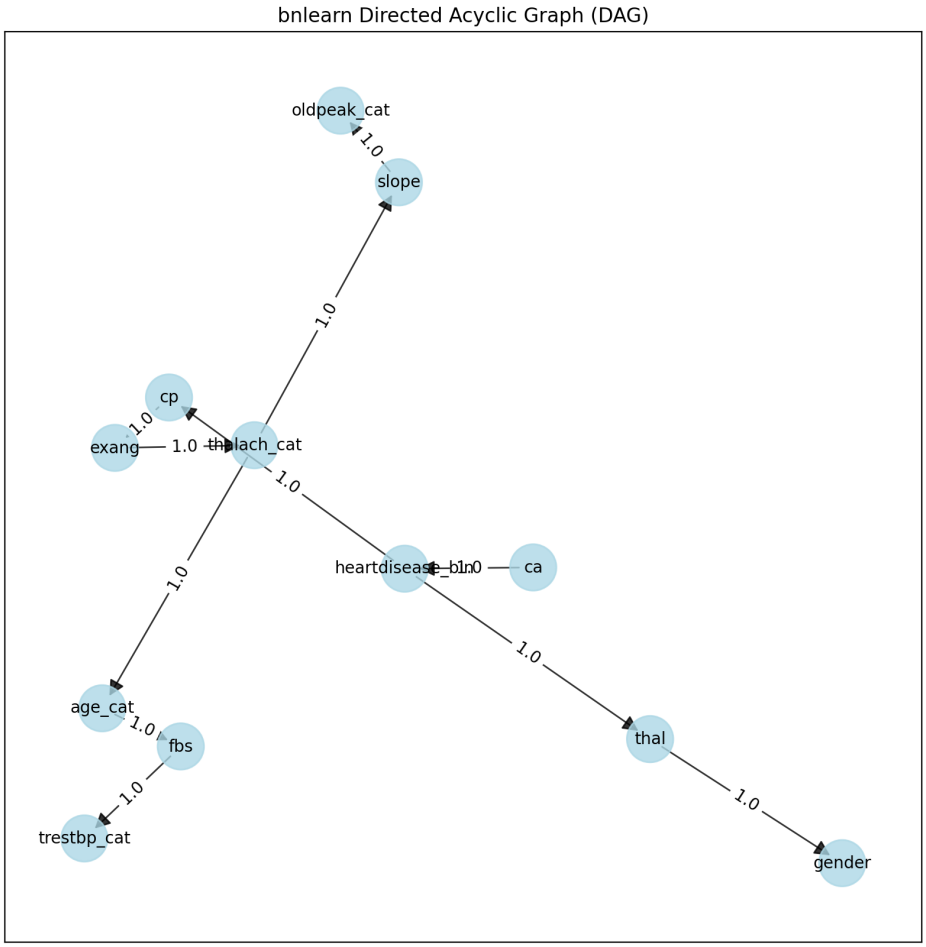


Figure 4: Data-driven DAG learned using Hill Climbing and BIC

Note. The learned network was estimated from the training data only. Arrow orientation is interpreted as part of the selected probabilistic factorisation. It is not treated as evidence of clinical causality.

4.3 Structural comparison

The expert and learned networks show partial structural agreement. Three directed edges are identical, five adjacencies are shared when direction is ignored, and two relationships are reversed.

Table 3: Structural comparison of the expert and learned DAGs

Comparison	Count	Interpretation / examples
Expert directed edges	16	Domain-informed network with four direct target parents
Learned directed edges	11	Sparser score-based structure
Common directed edges	3	heartdisease_bin → cp; heartdisease_bin → thal; slope → oldpeak_cat
Common adjacencies	5	Same variable pair linked regardless of direction
Reversed relationships	2	age_cat → thalach_cat vs. thalach_cat → age_cat; heartdisease_bin → ca vs. ca → heartdisease_bin
Learned-only examples	–	age_cat → fbs; fbs → trestbp_cat; cp → exang; thal → gender

The reversed age–heart-rate relationship illustrates why a score-based observational DAG should not be interpreted as causal. Likewise, the learned edge `thal` → `gender` is analytically useful as evidence of an empirical dependency selected by BIC, but it is not a plausible causal statement. The learned model is therefore evaluated primarily as a probabilistic and predictive structure.

5 Parameter Learning and CPT Interpretation

Both DAGs were parameterised using the training data only. Bayesian parameter learning with BDeu smoothing was used to estimate the conditional probability distributions. This was preferred to unsmoothed maximum-likelihood estimates because the dataset is small and some parent-state configurations are sparsely observed.

5.1 Expert target CPT

The expert target CPT conditions heart-disease probability on `age_cat`, `gender`, `trestbp_cat`, and `chol_cat`. The highest and lowest target-parent configurations are summarised below.

Table 4: Key expert-DAG target CPT patterns

Pattern	Condition	P(disease=1)
Highest fitted target-parent configuration	age=High, chol=High, gender=1, trestbps=High	0.622
Lowest fitted target-parent configuration	age=Low, chol=Low, gender=0, trestbps=Low	0.336
Largest gender swing	$0 \rightarrow 1$, other target parents fixed	$0.402 \rightarrow 0.622$
Largest age-category swing	Low $\rightarrow$ Medium, other target parents fixed	$0.425 \rightarrow 0.613$
Largest cholesterol-category swing	Medium $\rightarrow$ High, other target parents fixed	$0.436 \rightarrow 0.622$
Largest resting-BP-category swing	High $\rightarrow$ Low in one fixed configuration	$0.436 \rightarrow 0.622$

The first five patterns are directionally interpretable within the fitted model, but the resting-blood-pressure pattern is not monotonically clinical. It should therefore be reported as a local conditional CPT result rather than as a causal blood-pressure effect.

5.2 Direct-parent CPTs

The direct parent distributions are shown in Table 5. Resting blood pressure is itself conditioned on age in the expert graph, whereas age, gender, and cholesterol have marginal CPDs.

Table 5: CPT/CPD summaries for direct parents of the expert target

Node	Fitted probability distribution
age_cat	High 0.327; Low 0.338; Medium 0.335
gender	State 0: 0.465; State 1: 0.535
chol_cat	High 0.334; Low 0.336; Medium 0.330
trestbp_cat | age_cat	For age=High: High 0.362, Low 0.323, Medium 0.315; for age=Low: High 0.306, Low 0.365, Medium 0.329; for age=Medium: High 0.334, Low 0.320, Medium 0.346

5.3 Learned target CPT

In the learned DAG, `ca` is the only direct parent of the target.

The fitted probabilities were:

$$P(D = 1 \mid ca = 0) = 0.417, \quad P(D = 1 \mid ca = 1) = 0.531,$$
$$P(D = 1 \mid ca = 2) = 0.530, \quad P(D = 1 \mid ca = 3) = 0.519.$$

This produces a much simpler target CPT than the expert network, although inference can still incorporate evidence from other variables through the broader learned graph.

6 Probabilistic Inference

Inference was conducted in both fitted networks using clinically relevant evidence variables: gender, chest-pain type (`cp`), exercise-induced angina (`exang`), `thal`, discretised oldpeak and maximum heart rate, slope, and `ca`. All possible state combinations of these variables were enumerated, yielding 5,184 evidence profiles per model. Both networks successfully resolved all 5,184 profiles.

6.1 Highest- and lowest-probability evidence combinations

The expert network produced a wider range of posterior probabilities than the data-driven network. Its maximum posterior was approximately 0.793 and its minimum was approximately 0.192. The learned DAG ranged from approximately 0.307 to 0.670.

The highest-ranked profiles in both networks repeatedly included `gender`=1, `cp`=4, `thal`=7, and `ca`=1. This cross-model agreement is useful because it shows that the two different graph structures still identify similar high-posterior evidence patterns.

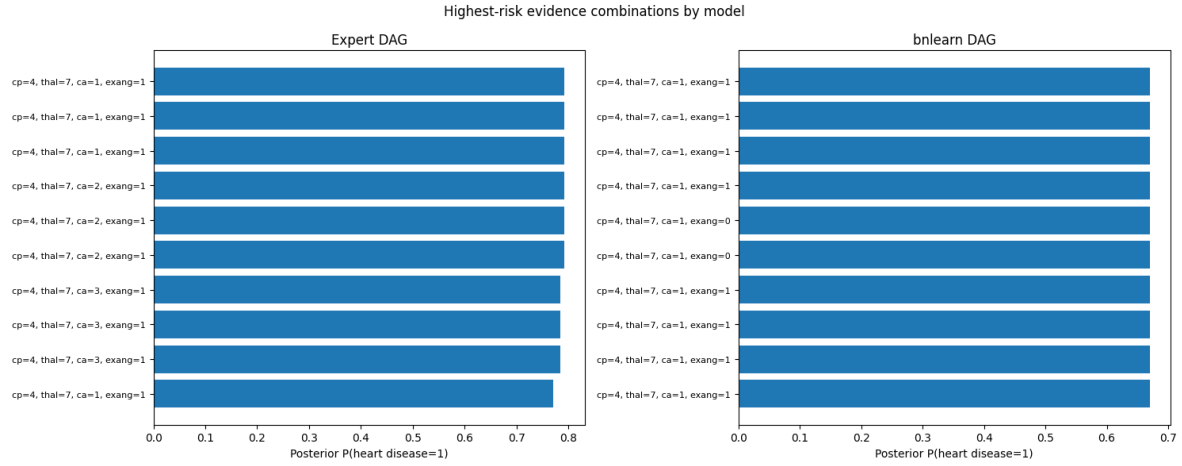


Figure 5: Highest-posterior evidence combinations by model

Note. The figure displays the top-ranked combinations among 5,184 profiles evaluated in each network. The plotted labels abbreviate the full evidence profiles; ranking is based on posterior P(heart disease=1).

Across all profiles, the mean absolute difference between the expert and learned posterior probabilities was 0.0564. Their Spearman rank correlation was 0.7664, indicating meaningful but incomplete agreement in how the models order evidence profiles.

Table 6: Inference comparison across the full evidence grid

Statistic	Expert DAG	bnlearn DAG
Evidence combinations resolved	5,184	5,184
Maximum posterior P(disease=1)	0.793	0.670
Minimum posterior P(disease=1)	0.192	0.307
Posterior range	0.601	0.363
Mean absolute posterior difference between models: 0.0564		
Spearman rank correlation between posterior rankings: 0.7664		

6.2 One-variable sensitivity analysis

To isolate the effect of changing a single evidence variable in the fitted models, `ca` was varied from 0 to 3 while all other evidence was held fixed. The largest shift occurred from `ca=0` to `ca=1`: the expert posterior increased from 0.708 to 0.793, while the learned posterior increased from 0.561 to 0.670.

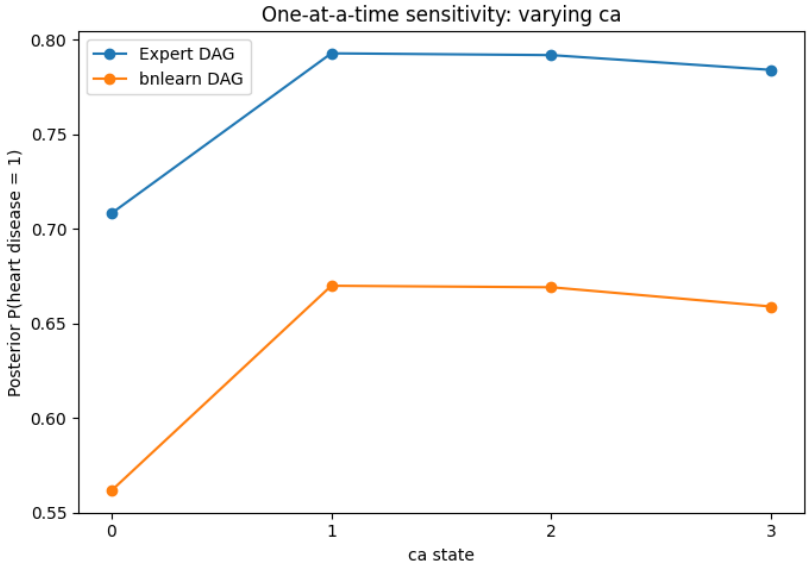


Figure 6: One-at-a-time posterior sensitivity to `ca`

Note. Only the `ca` state is varied; all other evidence is held fixed. The result is a sensitivity analysis of the fitted Bayesian networks and should not be interpreted as the effect of a causal intervention.

6.3 Reverse inference

The assignment also requires reverse inference after setting heart disease to 1. In both models, `cp=4` had the largest posterior among chest-pain states (approximately 0.340), and `thal=7` had the largest posterior among the three `thal` states (approximately 0.394). The expert network placed slightly more posterior mass on `exang=1` (0.507), whereas the learned network placed slightly more mass on `exang=0` (0.530). For `ca`, state 0 had the largest single posterior (approximately 0.272), but probability remained distributed across all four states.

These reverse-inference results answer which explanatory states become more probable once

disease is fixed as present. They do not imply that the corresponding states independently cause heart disease.

7 Model Evaluation and Reflection

Both fitted networks were evaluated on the same held-out test set of 61 observations, comprising 33 negative and 28 positive cases. The Expert DAG achieved the strongest overall predictive performance, with an accuracy of 90.2%, a balanced accuracy of 90.1%, an F1-score of 0.893, an ROC-AUC of 0.961, and a PR-AUC of 0.953. The data-driven bnlearn DAG also performed strongly, achieving 85.3% accuracy, a balanced accuracy of 85.0%, an F1-score of 0.836, an ROC-AUC of 0.915, and a PR-AUC of 0.909. By comparison, the majority-class baseline achieved only 54.1% accuracy and a balanced accuracy of 50.0%, while failing to identify any positive heart-disease cases. These results indicate that both Bayesian networks substantially outperform the no-skill benchmark, with the Expert DAG providing the stronger discrimination and overall classification performance in the current held-out sample.

Table 7: Held-out classification performance

Model	Accuracy	Bal. Acc.	Precision	Recall	Specificity	F1	ROC-AUC	PR-AUC
Majority baseline	0.541	0.500	0.000	0.000	1.000	0.000	0.500	0.459
Expert DAG	**0.902**	**0.901**	**0.893**	**0.893**	**0.909**	**0.893**	**0.961**	**0.953**
bnlearn DAG	0.853	0.850	0.852	0.821	0.879	0.836	0.915	0.909

The confusion matrices provide a direct view of the classification errors. The majority baseline classifies every case as no disease, correctly identifying 33 negatives but missing all 28 positives. The expert DAG records 30 true negatives, 25 true positives, three false positives, and three false negatives. The learned DAG records 29 true negatives, 23 true positives, four false positives, and five false negatives.

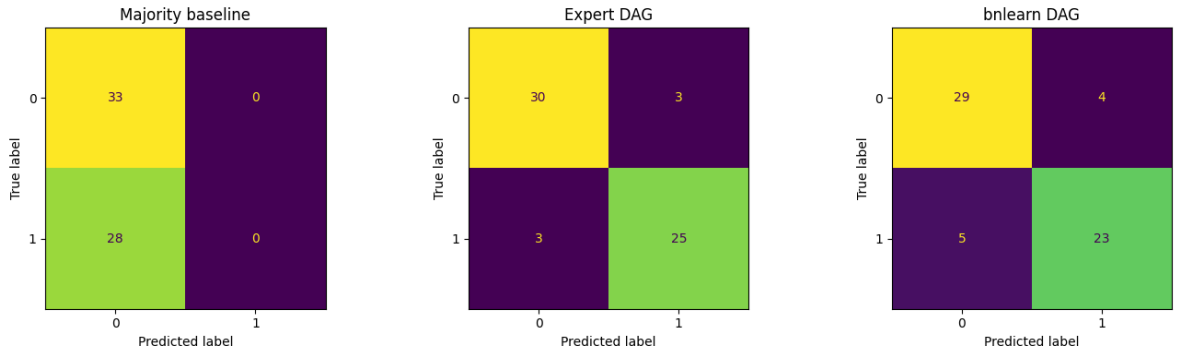
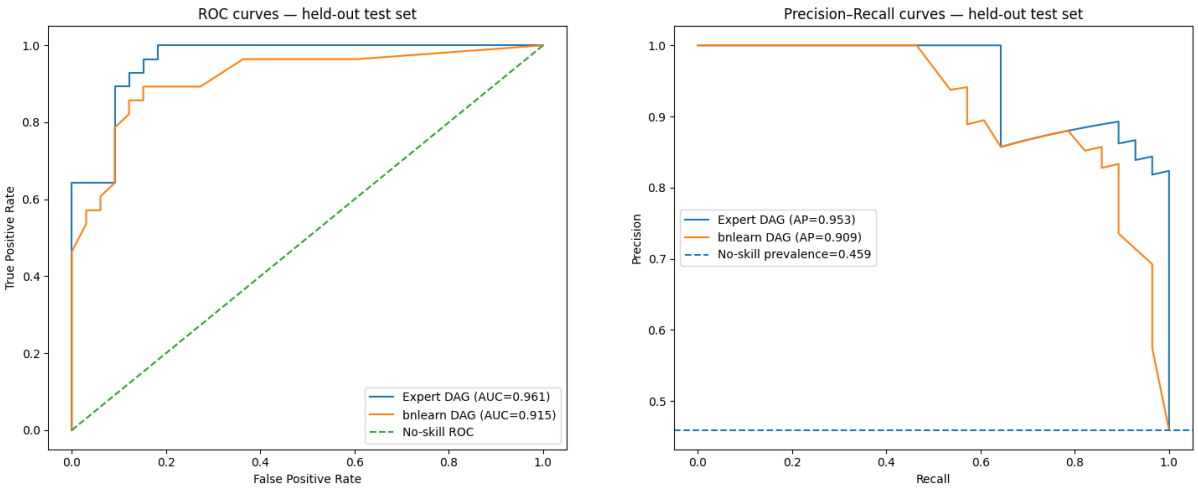


Figure 7: Confusion matrices on the held-out test set

Note. Rows indicate true labels and columns indicate predicted labels. The test set contains 61 observations: 33 negatives and 28 positives.

The ROC and precision–recall curves confirm the same ranking. The expert network achieves ROC-AUC of 0.961 and PR-AUC of 0.953, compared with 0.915 and 0.909 for the learned network.



(a) ROC curves

(b) Precision–recall curves

Figure 8: Held-out discrimination performance

Note. The ROC reference line represents no-skill discrimination. The precision–recall reference line equals the disease prevalence in the held-out sample. Both curves favour the expert DAG in the current split.

7.1 Predictive performance versus interpretability

The expert DAG is the preferred model in this assignment for two reasons. First, it produces the strongest held-out predictive metrics. Second, its edges were explicitly justified and organised into interpretable pathways. The learned network is structurally sparser (11 edges versus 16), but some learned directions, for example `thal` → `gender` and `thalach_cat` → `age_cat`—are difficult to defend as domain structure and are therefore treated as empirical dependencies rather than causal mechanisms.

The finding that the expert graph performs better should nevertheless be interpreted cautiously. The test set contains only 61 observations, and performance reflects one stratified split. The difference, therefore, provides evidence in favour of the expert DAG for this analysis, not proof of superior population performance.

8 Conclusion

This study constructed and evaluated two Bayesian networks for binary heart-disease prediction using the Cleveland dataset. The complete workflow incorporated data-quality checks, training-only preprocessing, expert and data-driven structure learning, Bayesian parameter estimation, detailed probabilistic inference, and held-out classification evaluation.

The expert and learned networks did not produce identical structures, but they showed meaningful overlap in both graph adjacency and inference rankings. Their posterior probabilities differed by 0.056 on average across 5,184 evidence combinations, with a rank correlation of 0.766. Both networks identified broadly similar high-posterior evidence patterns, particularly profiles involving `cp=4`, `thal=7`, and `ca=1`.

The expert DAG was the stronger model on the held-out test set, achieving an F1-score of 0.893 and ROC-AUC of 0.961, compared with 0.836 and 0.915 for the data-driven DAG. It also

provided clearer domain-informed pathways. Accordingly, the expert DAG is selected as the preferred model for this assignment.

The result should not be interpreted as evidence that the expert structure is medically causal or ready for clinical deployment. It indicates that, within this dataset and split, the domain-informed Bayesian network provides a favourable combination of predictive performance and interpretability.

9 Limitations

Several limitations qualify the findings.

- The total sample contains only 303 observations, and the held-out test set contains 61 observations. Performance estimates may therefore be sensitive to the particular split.
- Continuous clinical variables were discretised into three states. This makes CPT estimation and interpretation tractable but sacrifices within-bin information.
- Missing values in `ca` and `thal` were handled using simple training-mode imputation. More sophisticated missing-data methods could be explored in larger studies.
- The expert DAG reflects a defensible modelling structure for the assignment, but it was not independently reviewed or validated as a clinical causal graph.
- Hill Climbing with BIC can converge to a locally preferred structure, and different score-equivalent or near-equivalent graphs may represent similar dependency information.
- Model evaluation focuses primarily on discrimination and classification. Formal probability calibration was not assessed.

10 Recommendations

For the current assignment, the expert DAG should be retained as the primary network because it combines stronger held-out performance with clearer structural justification. The learned DAG remains valuable as an empirical benchmark and as a mechanism for identifying relationships that differ from the expert specification.

For future analytical development, the following extensions are recommended:

- use repeated cross-validation or bootstrap resampling to quantify uncertainty around model-performance differences;
- assess posterior calibration in addition to ROC-AUC and PR-AUC;
- obtain clinical review of expert edges before making any causal or diagnostic interpretation;
- examine alternative discretisation schemes and sensitivity to the BDeu prior;
- compare the Bayesian networks against additional transparent classification baselines; and
- if the model were ever considered for operational decision support, validate it on an external patient sample before use.

11 Reference List

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